INCH-POUND
MIL-DTL-16377/57C(SH)
30 June 2014
SUPERSEDING
MIL-DTL-16377/57B(SH)
02 August 1996

DETAIL SPECIFICATION SHEET

# FIXTURES, LIGHTING; FLUORESCENT AND SSL EQUIVALENT, GENERAL LIGHTING, 60, 80, AND 120 WATTS, 120 VOLTS, 60 HERTZ, DRIPPROOF, SYMBOLS 348, 349, 350, 348SSL, 349SSL, AND 350SSL

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-16377.

All dimensions are in inches. For tolerances other than specified herein, see 3.5.2 of MIL-DTL-16377.

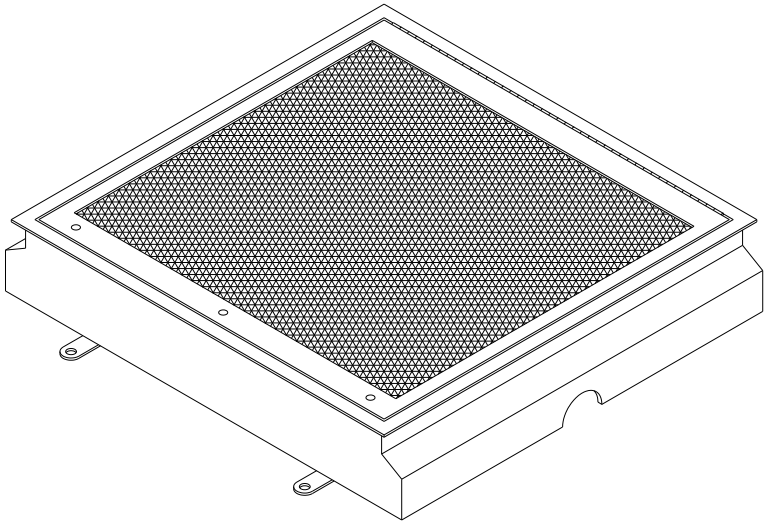
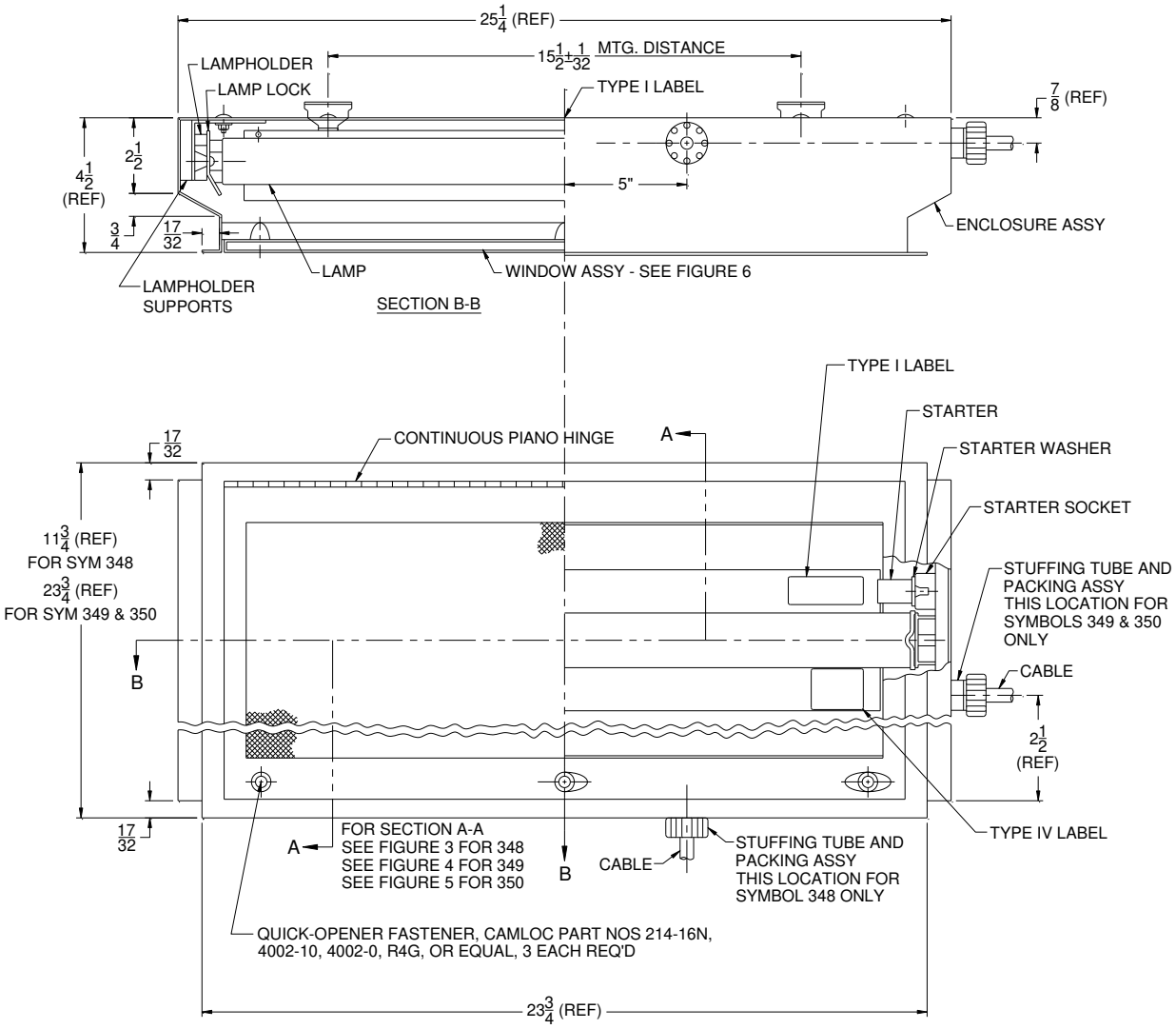


FIGURE 1. Typical view of symbols 349 and 350.



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NOTES: DIMENSIONS SHOWN MAY BE MODIFIED TO ACCOMMODATE BALLAST. CRITICAL DIMENSIONS (MOUNTING, WINDOW AREA, LAMP SPACING, ETC.) MUST BE MAINTAINED.

FIGURE 2. Typical view of symbols 348, 349, and 350.



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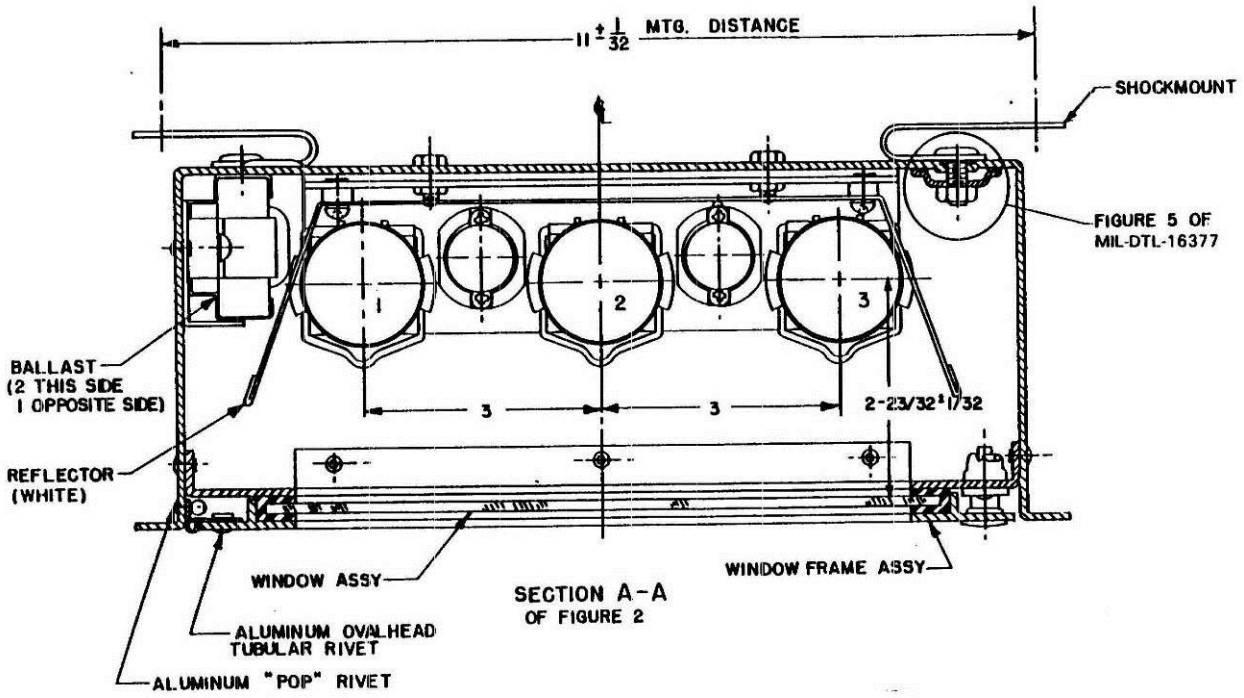



FIGURE 3. Typical section view of symbol 348.

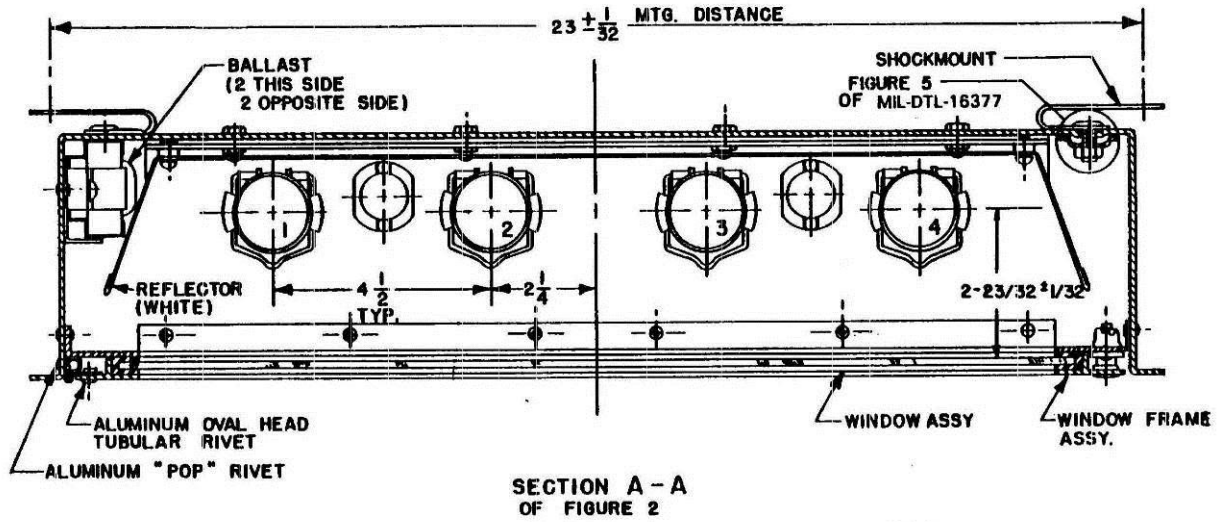
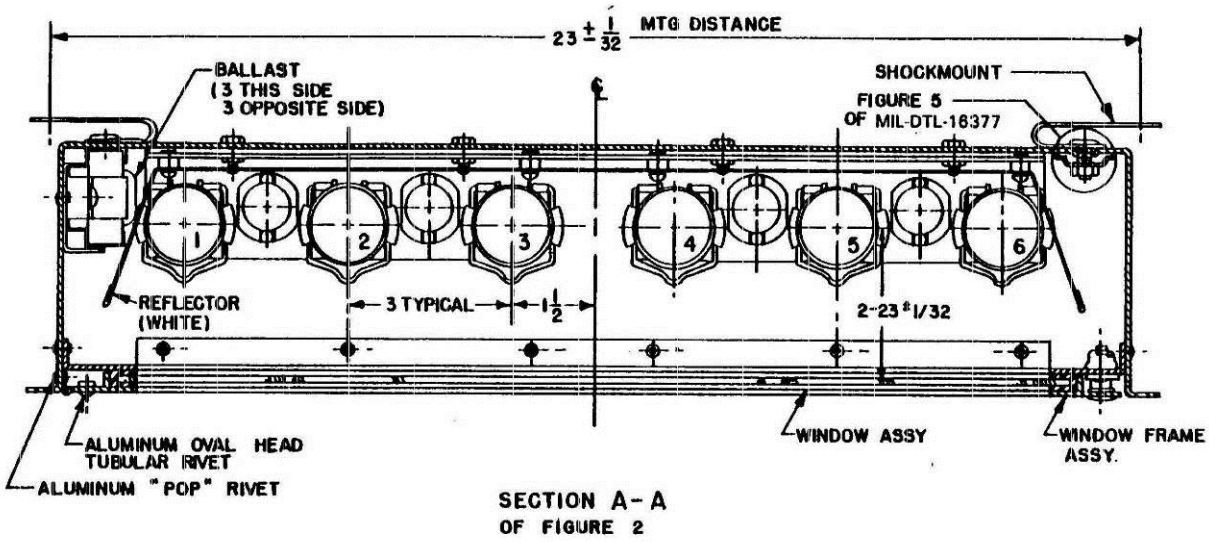



FIGURE 4. Typical section of symbol 349.



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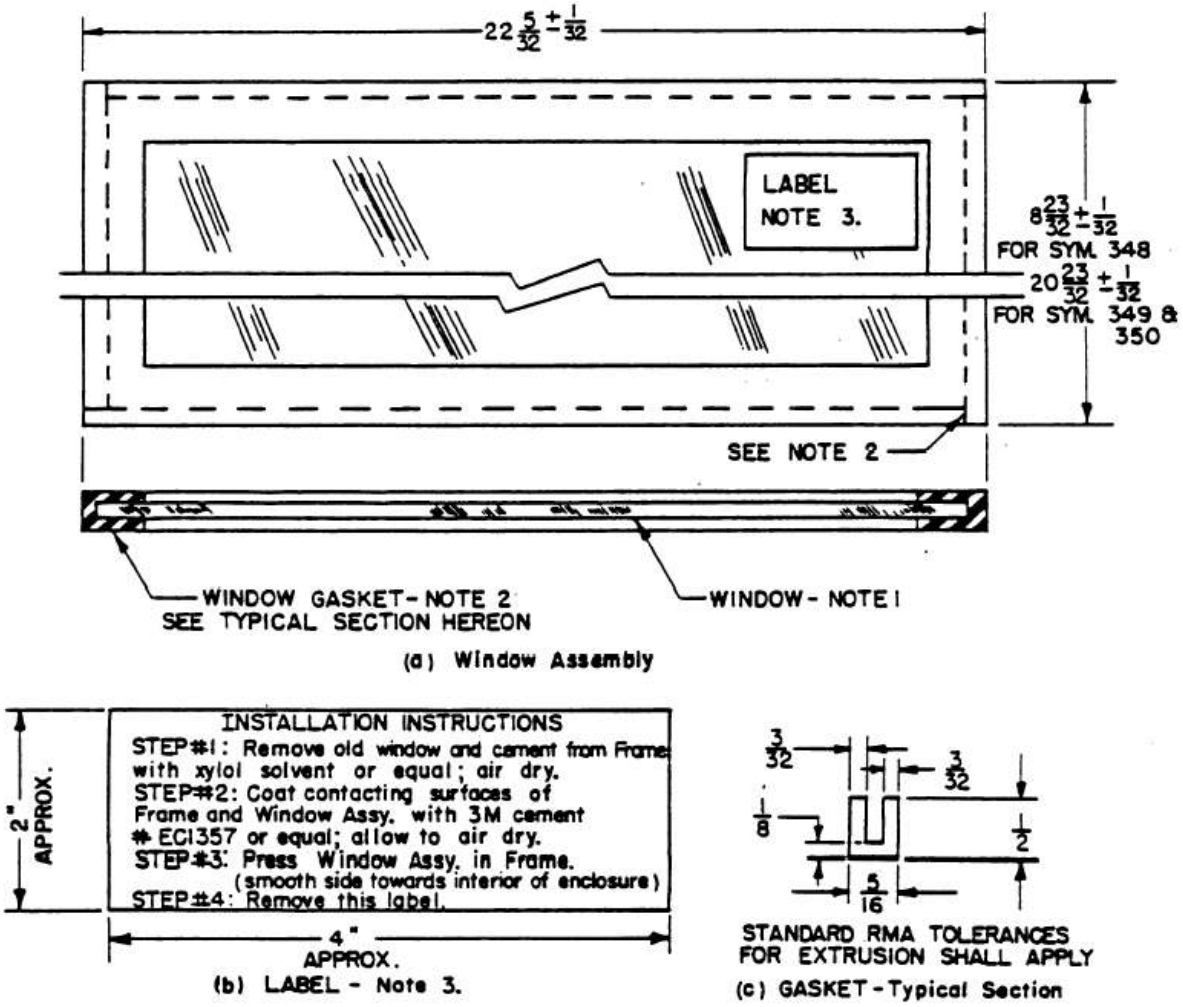



NOTE: For red and white illumination, installing activity shall provide red filter tubes on #2 and #5 lamps.

FIGURE 5. Typical section of symbol 350.



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NOTES:

1. Material for window shall be ½-inch thick, clear, prismatic (one side only), heat resistant, cast acrylic plastic sheet, Type CLEAR-2, of MIL-DTL-24191.
2. Material for window gasket shall be neoprene rubber 25-35 durometer. Gasket shall be cemented to window. Gasket may be furnished in segments.
3. For window assemblies furnished as repair parts, a Type IV label shall be furnished in accordance with MIL-DTL-16377. Location and the information to be shown on this label shall be as shown hereon.

FIGURE 6. Window assembly.



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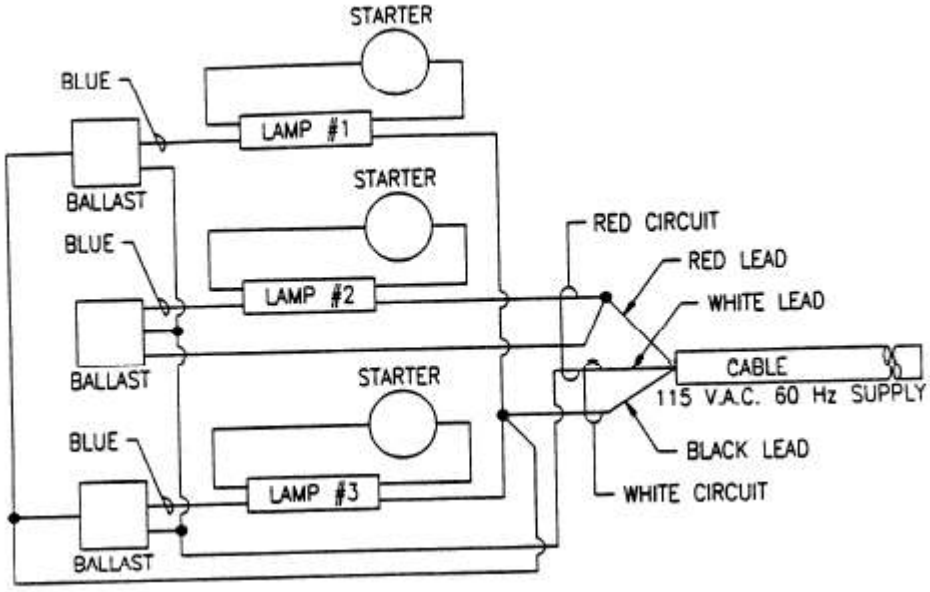



FIGURE 7. Wiring diagram for symbol 348.

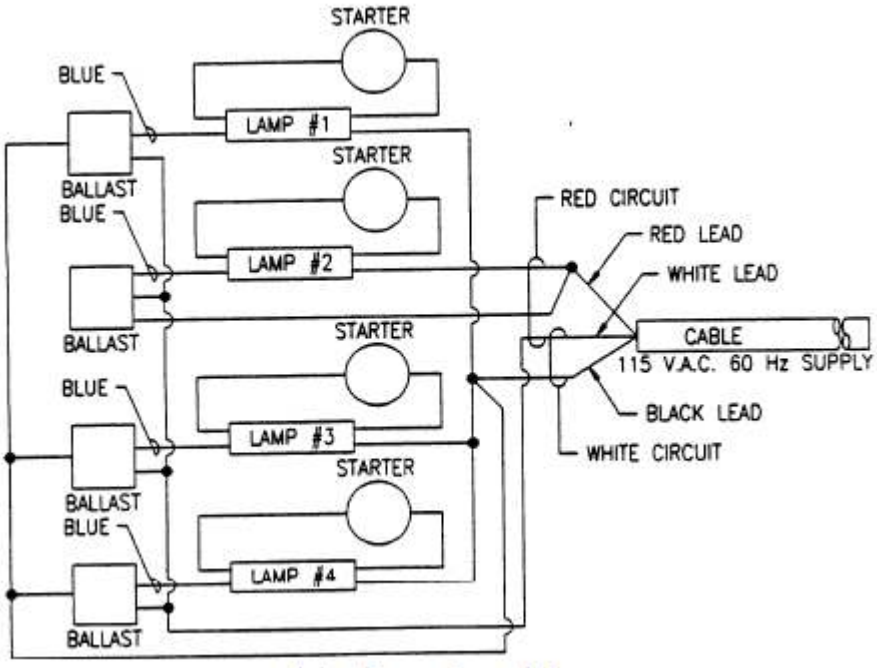



FIGURE 8. Wiring diagram for symbol 349.



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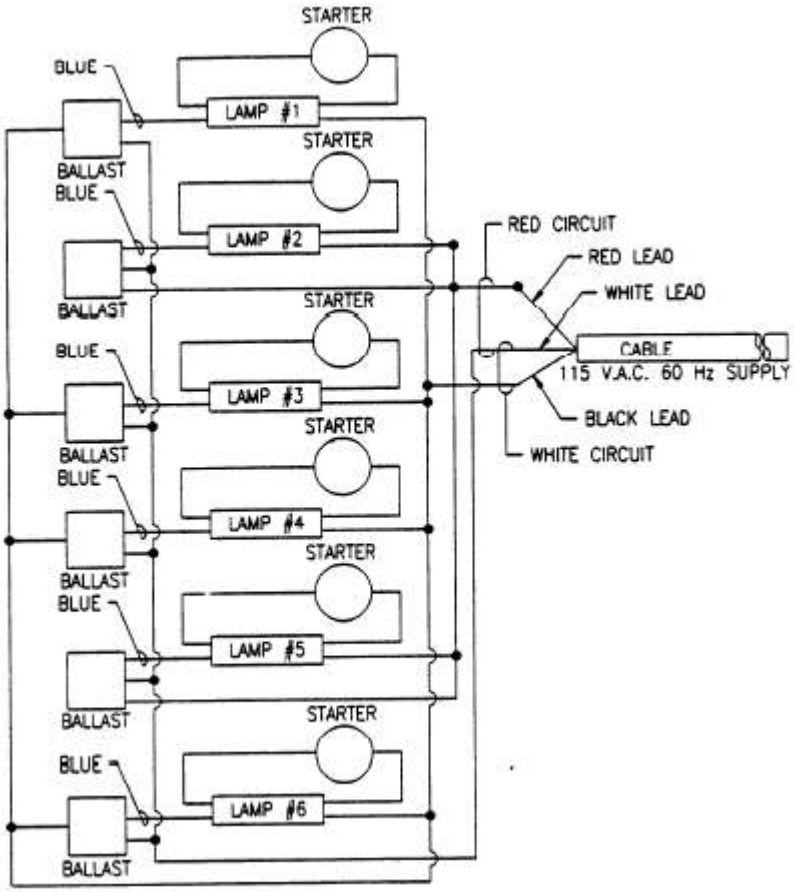
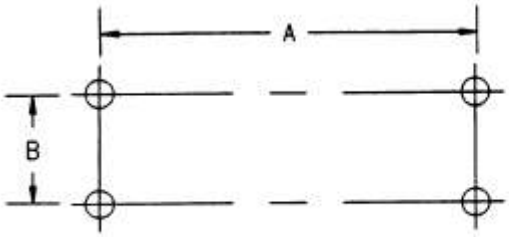



FIGURE 9. Wiring diagram for symbol 350.

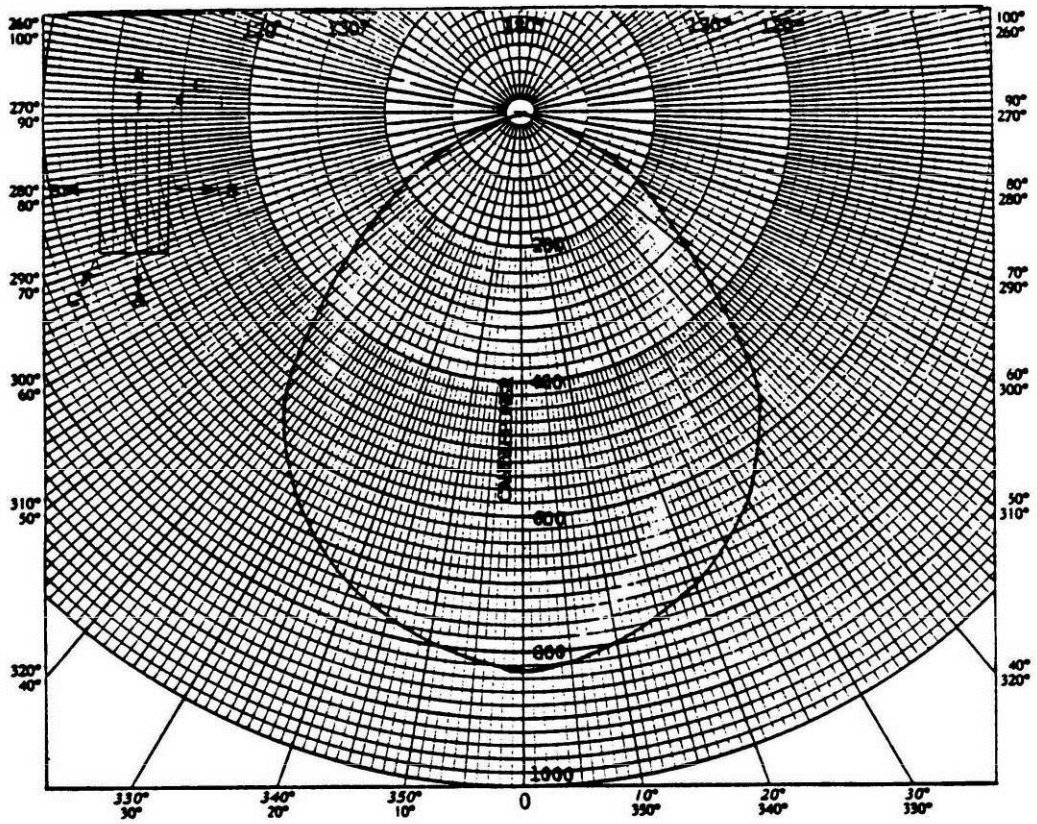



| SYMBOL | DIMENSION | |
|---|---|---|
| | A | B |
| 348 | 11 | 15.5 |
| 349, 350 | 23 | 15.5 |

FIGURE 10. Mounting dimensions.



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| ANGLE | CANDLEPOWER* SYMBOL 348 |
|---|---|
| 0 | 825 |
| 5 | 810 |
| 10 | 800 |
| 15 | 760 |
| 20 | 750 |
| 25 | 715 |
| 30 | 670 |
| 35 | 620 |
| 40 | 550 |
| 45 | 440 |
| 50 | 330 |
| 55 | 240 |
| 60 | 180 |

| ANGLE | CANDLEPOWER* SYMBOL 348 |
|---|---|
| 65 | 130 |
| 70 | 110 |
| 75 | 90 |
| 80 | 45 |
| 85 | 20 |
| 90 | 10 |
| 95 | 5 |
| 100 | 0 |
| 105 | 0 |
| 110 | 0 |
| 115 | 0 |
| 120 | 0 |
| 125 | 0 |

*Average values for planes A-A, B-B and C-C.

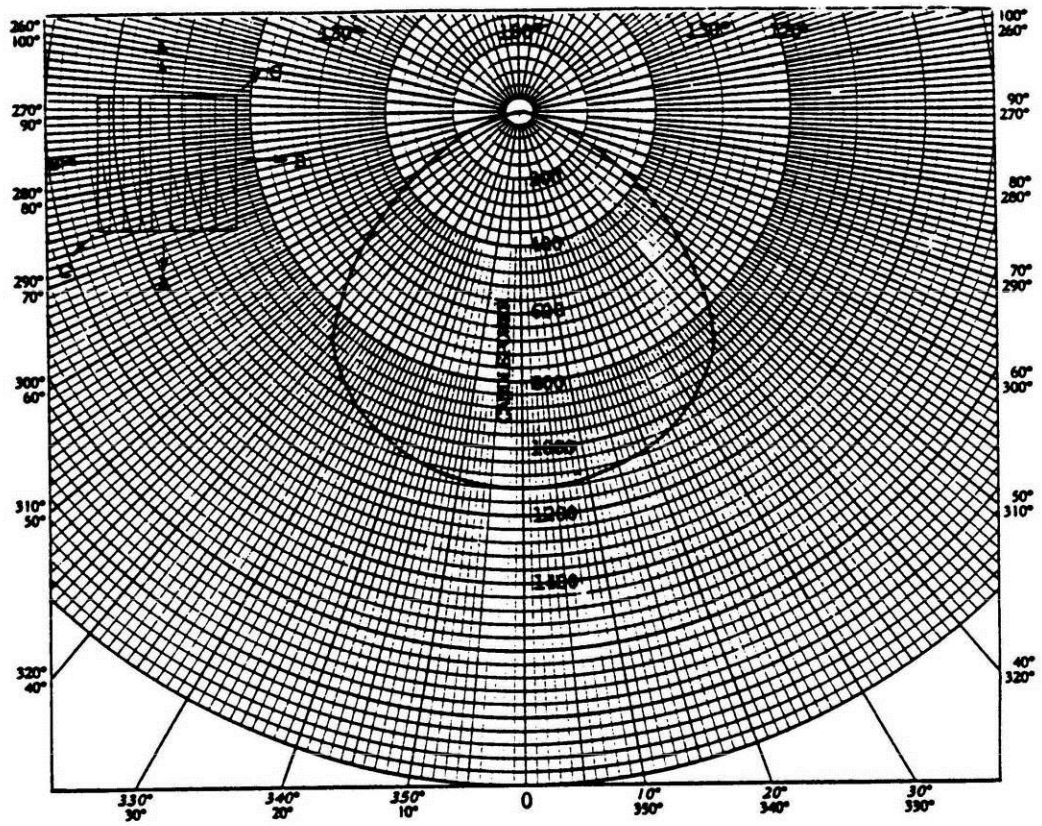
PHOTOMETRIC TEST DATA:

| | |
|---|---|
| Distance: | 10 feet. |
| Plane: | At right angles to long axis of fixture through geometric center of lamp. |
| Voltage: | 115 volts, 60 hertz. |
| Ambient temperature: | 76 °F (24.4 °C). |
| Lamps: | Use lamps furnished with fixtures seasoned for at least 100 hours. |
| Ballast: | Use ballast furnished with fixture. |
| Lamp correction: | Distribution curves based on 1,280 lumens per lamp. |

FIGURE 11. Candlepower distribution curves for symbol 348.



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| ANGLE | CANDLEPOWER* SYMBOL 349 |
|---|---|
| 0 | 1120 |
| 5 | 1120 |
| 10 | 1100 |
| 15 | 1080 |
| 20 | 1070 |
| 25 | 1035 |
| 30 | 1000 |
| 35 | 935 |
| 40 | 890 |
| 45 | 825 |
| 50 | 600 |
| 55 | 450 |
| 60 | 275 |

| ANGLE | CANDLEPOWER* SYMBOL 349 |
|---|---|
| 65 | 220 |
| 70 | 165 |
| 75 | 55 |
| 80 | 45 |
| 85 | 25 |
| 90 | 0 |
| 95 | 0 |
| 100 | 0 |
| 105 | 0 |
| 110 | 0 |
| 115 | 0 |
| 120 | 0 |
| 125 | 0 |

*Average values for planes A-A, B-B and C-C.

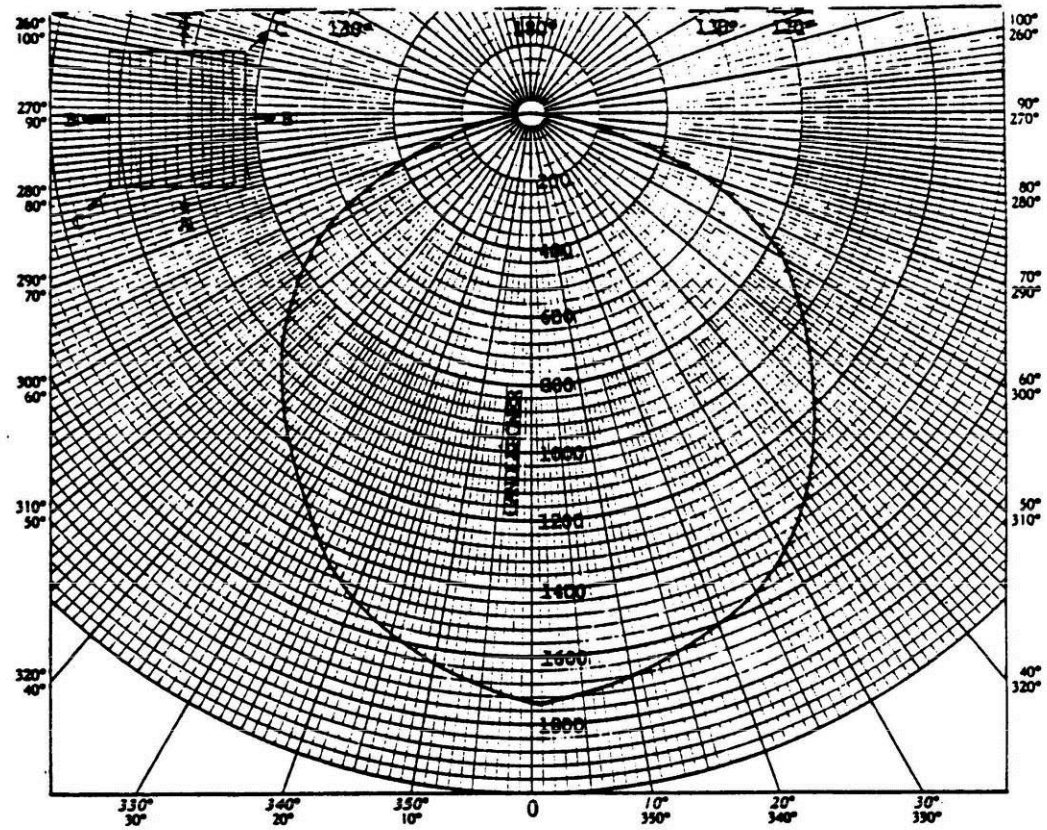
PHOTOMETRIC TEST DATA:

| | |
|---|---|
| Distance: | 10 feet. |
| Plane: | At right angles to long axis of fixture through geometric center of lamp. |
| Voltage: | 115 volts, 60 hertz. |
| Ambient temperature: | 76 °F (24.4 °C). |
| Lamps: | Use lamps furnished with fixtures seasoned for at least 100 hours. |
| Ballast: | Use ballast furnished with fixture. |
| Lamp correction: | Distribution curves based on 1,280 lumens per lamp. |

FIGURE 12. Candlepower distribution curves for symbol 349.



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| ANGLE | CANDLEPOWER* SYMBOL 350 |
|---|---|
| 0 | 1760 |
| 5 | 1760 |
| 10 | 1740 |
| 15 | 1720 |
| 20 | 1670 |
| 25 | 1630 |
| 30 | 1550 |
| 35 | 1475 |
| 40 | 1365 |
| 45 | 1255 |
| 50 | 880 |
| 55 | 660 |
| 60 | 440 |

| ANGLE | CANDLEPOWER* SYMBOL 350 |
|---|---|
| 65 | 330 |
| 70 | 220 |
| 75 | 110 |
| 80 | 55 |
| 85 | 25 |
| 90 | 0 |
| 95 | 0 |
| 100 | 0 |
| 105 | 0 |
| 110 | 0 |
| 115 | 0 |
| 120 | 0 |
| 125 | 0 |

*Average values for planes A-A, B-B and C-C.

PHOTOMETRIC TEST DATA:

| | |
|---|---|
| Distance: | 10 feet. |
| Plane: | At right angles to long axis of fixture through geometric center of lamp. |
| Voltage: | 115 volts, 60 hertz. |
| Ambient temperature: | 76 °F (24.4 °C). |
| Lamps: | Use lamps furnished with fixtures seasoned for at least 100 hours. |
| Ballast: | Use ballast furnished with fixture. |
| Lamp correction: | Distribution curves based on 1,280 lumens per lamp. |

FIGURE 13. Candlepower distribution curves for symbol 350.



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REQUIREMENTS:

CLASSIFICATION:

Type: I
Class: 2

SYMBOL NUMBERS:

348: Complete lighting fixture assembly, flush mounting, with three 20-watt cool white fluorescent lamps and clear prismatic window.

349: Complete lighting fixture assembly, flush mounting, with four 20-watt cool white fluorescent lamps and clear prismatic window.

350: Complete lighting fixture assembly, flush mounting, with six 20-watt cool white fluorescent lamps and clear prismatic window.

DESIGN:

Electrical rating: 51, 68, and 102 watts, 120 volts, 60 hertz.
Dimension and configurations: See figures 1 through 5.
Distribution curves: Candlepower. See figures 11, 12, and 13.
Cable: 10 feet of Type LSTHOF-3 in accordance with MIL-DTL-24643/3.
Wiring diagram: See figures 7, 8, and 9.
Window assembly: See figure 6.
Mounting dimensions: See figure 10.
Enclosure: Dripproof 15 degrees.
Weight:
Symbol 348: 20 pounds maximum.
Symbol 349: 26 pounds maximum.
Symbol 350: 30 pounds maximum.
Designation and marking: Labels shall be furnished in accordance with the following types of MIL-DTL-16377:

a. Two Type I.

b. One Type IV label shall be applied as shown on figure 2. The information to be shown on the Type IV label shall be “USE THE FOLLOWING REPAIR PARTS ONLY:” followed by a list of all standard parts shown in table I by name and stock number only.

Standard parts: The standard parts shown in table I shall be used in the construction of the fixtures specified herein.



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REQUIREMENTS:

CLASSIFICATION:

Type: III
Class: 2

SYMBOL NUMBERS:

348SSL: Complete lighting fixture assembly, flush mounting, with LED light source and clear prismatic window.
349SSL: Complete lighting fixture assembly, flush mounting, with LED light source and clear prismatic window.
350SSL: Complete lighting fixture assembly, flush mounting, with LED light source and clear prismatic window.

DESIGN:

Electrical rating:
- Symbol 348SSL: 60 watts maximum, 120 volts, 60 hertz.
- Symbol 349SSL: 80 watts maximum, 120 volts, 60 hertz.
- Symbol 350SSL: 120 watts maximum, 120 volts, 60 hertz.

Dimension and configurations: External dimensions cannot exceed those specified on figures 1 through 5.
Photometric data: See figures 11, 12, or 13, as applicable, and 3.7.1 of MIL-DTL-16377.
Chromaticity: See 3.7.4 of MIL-DTL-16377.
Fail-safe circuit: See 3.17 of MIL-DTL-16377.
Cable: 10 feet of Type LSTHOF-3 in accordance with MIL-DTL-24643/3.
Wiring: Shall maintain same control scheme for light output as legacy fluorescent fixture. Refer to figures 7 through 9, as applicable.
Window assembly: See figure 6.
Mounting dimensions: See figure 10.
Enclosure: Dripproof 15 degrees.
Weight:
- Symbol 348SSL: 20 pounds maximum.
- Symbol 349SSL: 26 pounds maximum.
- Symbol 350SSL: 30 pounds maximum.

Designation and marking: Labels shall be furnished in accordance with the following types of MIL-DTL-16377:

a. Two Type I.
b. One Type IV shall be applied as shown on figure 2. The information to be shown on the Type IV label shall be “USE THE FOLLOWING REPAIR PARTS ONLY:” followed by:
   (1) A list of all standard parts shown in table I and window assemblies of table V by name and stock number only, if appropriate, or
   (2) A list of available replacement parts and stock numbers, or if unavailable, manufacturer’s part numbers.

Standard parts: The standard parts shown in table I shall be used in the construction of the fixtures specified herein.



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EXAMINATION AND TESTS:

Examination and tests shall be as specified in MIL-DTL-16377 and as follows:

First article inspection:
- Complete fixtures: See table II for Type I, table III for Type III.
- Window assemblies: See table IV.

Conformance:
- Comparison inspection: Same as first article inspection.
- Conformance inspection: See MIL-DTL-16377.

GENERAL INFORMATION:

Assignment of military part numbers and general information is shown in table V.

TABLE I. Standard parts.

| Name | Specification or Military Part Number | National Stock Number (NSN) | Remarks |
|---|---|---|---|
| Lamp | A-A-55005-1 | 6240-00-152-2996 | Industry number F20T12/CW |
| Lamp lock | M16377/40-002 | 6250-00-399-7040 | |
| Lampholder | UL 496 | 6250-00-284-0481 | SUPP. SB, FIG. SB12 |
| Startersocket | UL 496 | 6250-00-295-2738 | SUPP. SB, FIG. SB14 |
| Starter | NEMA ANSI C78.180 | 6250-00-299-2884 | Commercial designation: FS2 |
| Ballast | M16377/72-001 | 6250-01-465-6716 | |
| Shock mount:<br>Symbol 348<br>Symbols 349, 350 | <br>M16377/41-009 or<br>M16377/41-008 | <br>6210-00-973-1747 | <br>See 3.5.16 of MIL-DTL-16377 |
| Red filter tube | M16377/42-003 | 6210-00-295-2838 | Not furnished, see 3.7.9 of MIL-DTL-16377 |
| Broad band blue filter tube | M16377/42-005 | 6210-00-441-5138 | Not furnished, see 3.7.9 of MIL-DTL-16377 |
| Amber filter tube | M16377/42-006 | 6210-01-166-7184 | Not furnished, see 3.7.9 of MIL-DTL-16377 |
| Gray filter tube (1% transmission) | M16377/42-007 | 6210-01-498-8001 | Not furnished, see 3.7.9 of MIL-DTL-16377 |
| Gray filter tube (2.5% transmission) | M16377/42-009 | 6210-01-313-8168 | Not furnished, see 3.7.9 of MIL-DTL-16377 |



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TABLE II. First article and comparison inspection for complete type I fixtures. [1/]

| Test | Requirement | Test Method | Remarks |
|---|---|---|---|
| Examination | 3.13 | 4.5 | |
| Operation | 3.13 | 4.8.1 | |
| Dielectric withstanding voltage | 3.6.15 | 4.8.2 | [2/] |
| Insulation resistance | 3.6.16 | 4.8.3 | [2/] |
| Enclosure effectiveness | 3.5.11 | 4.8.14.2 | [2/, 3/] |
| Shock | 3.5.12 | 4.8.7 | [2/] |
| Vibration | 3.5.13 | 4.8.8 | [2/] |
| Enclosure effectiveness | 3.5.11 | 4.8.14.4 | [2/, 3/] |
| Dielectric withstanding voltage | 3.6.15 | 4.8.2 | [2/] |
| Insulation resistance | 3.6.16 | 4.8.3 | [2/] |
| Noise | 3.5.14 | 4.8.9 | |
| Electromagnetic interference | 3.6.14 | 4.8.16 | |
| Continuity of grounding | 3.6.13.4 | 4.8.12 | |
| Optical uniformity | 3.7.6.1.1 | 4.8.4 | |
| Light output | 3.7.1 | 4.8.6 | Candlepower |
| Magnetic permeability | 3.4.8 | 4.8.18 | |

NOTES:

[1/] Requirement and test method paragraph numbers refer to MIL-DTL-16377.

[2/] In accordance with 4.3.1 of MIL-DTL-16377, tests shall be performed on one luminaire in the order specified.

[3/] Dripproof 15 degrees.



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TABLE III. First article and comparison inspection for complete type III fixtures. [1/]

| Test | Requirement | Test Method | Remarks |
|---|---|---|---|
| Examination | 3.13 | 4.5 | |
| Operation | 3.13 | 4.8.1 | |
| Power interface | 3.6.18 | 4.8.21 | |
| Power quality | 3.6.18.1 | 4.8.22 | |
| Voltage transient spike | 3.6.18.2 | 4.8.23 | |
| Electromagnetic interference | 3.6.14 | 4.8.16 | |
| Dielectric withstanding voltage | 3.6.15 | 4.8.2 | 2/ |
| Insulation resistance | 3.6.16 | 4.8.3 | 2/ |
| Enclosure effectiveness | 3.5.11 | 4.8.14 | 2/, 3/ |
| Shock | 3.5.12 | 4.8.7 | 2/ |
| Vibration | 3.5.13 | 4.8.8 | 2/ |
| Enclosure effectiveness | 3.5.11 | 4.8.14 | 2/, 3/ |
| Dielectric withstanding voltage | 3.6.15 | 4.8.2 | 2/ |
| Insulation resistance | 3.6.14 | 4.8.3 | 2/ |
| Noise | 3.5.14 | 4.8.9 | 2/ |
| Continuity of grounding | 3.6.13.4 | 4.8.12 | |
| Photometrics | 3.7.1 | 4.8.6 | Candlepower |
| Brightness | 3.7.5 | 4.8.6.4 | |
| Chromaticity | 3.7.4 | 4.8.6.5 | |
| Correlated color temperature (CCT) | 3.7.4.1.1 | 4.8.6.6 | |
| Lumen maintenance | 3.7.11 | 4.8.20 | |
| Fail-safe circuit design | 3.17 | 4.8.24 | |
| Optical uniformity | 3.7.7.1 | 4.8.4 | |
| Stress relief | 3.8.5.3 | 4.8.5 | |
| Magnetic permeability | 3.4.8 | 4.8.18 | |

NOTES:

1/ Requirement and test method paragraph numbers refer to MIL-DTL-16377.

2/ In accordance with 4.3.1 of MIL-DTL-16377, tests shall be performed on one luminaire in the order specified.

3/ Dripproof, 15 degrees.



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TABLE IV. First article and comparison inspection for window assemblies. 1/

| Test | Requirement | Test Method |
|---|---|---|
| Examination | 3.13 | 4.5 |
| Optical uniformity | 3.7.6.1.1 | 4.8.4 |
| NOTE:<br>1/ Requirement and test method paragraph numbers refer to MIL-DTL-16377. | | |

TABLE V. Part number and general information.

| Military Part Number M16377/57- | Item Name | National Stock Number (NSN) | Remarks |
|---|---|---|---|
| 348 | Complete fixture, Symbol 348 | 6210-00-115-9186 | |
| 349 | Complete fixture, Symbol 349 | 6210-00-115-9152 | |
| 350 | Complete fixture, Symbol 350 | 6210-00-115-9150 | |
| 348SSL | Complete fixture, solid state, Symbol 348SSL | | |
| 349SSL | Complete fixture, solid state, Symbol 349SSL | | |
| 350SSL | Complete fixture, solid state, Symbol 350SSL | | |
| 001 | Window assembly for Symbol 348 | 6210-00-115-9188 | figure 6 |
| 002 | Window assembly for Symbols 349 and 350 | 6210-00-115-9189 | figure 6 |

CHANGES FROM PREVIOUS ISSUE: Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodian:
Navy – SH

Review activity:
DLA – IS

Preparing activity:
Navy – SH
(Project 6210-2012-032)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.



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